

Appendix S2

Detecting disease progression from animal movement using hidden Markov models

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Ecology

Section S1. Choosing initial parameter values for the estimation

To avoid convergence issues in the likelihood optimization, we fit 15 models per state model (i.e., 3, 4, and 5 state HMMs) with different sets of starting values and select the best model fit among those based on log-likelihood values.

The best model per state is listed below with their maximum log-likelihood value among 15 model runs:

3-state HMM:

Maximum log-likelihood values:

```
-833747.1 -822614.4 -822614.4 -822614.4 -833747.1 -822614.4 -822614.4
-822614.4 -822614.4 -822614.4 -822614.4 -822614.4 -822614.4 -822614.4
-822614.4
```

Best model:

Value of the maximum log-likelihood: -822614.4

step parameters:

```
      state 1  state 2  state 3
mean 4.226161 317.6875 1809.043
sd   3.159653 364.1812 1064.134
```

angle parameters:

```
              state 1    state 2    state 3
mean          3.1383642 0.07831160 -0.0368014
concentration 0.4200796 0.08711126  1.5305686
```

4-state HMM:

Maximum log-likelihood values:

```
-819189.6 -819189.6 -819189.6 -819189.6 -819189.6 -819189.6 -819189.7
-819189.6 -819189.7 -819189.6 -819189.6 -819189.6 -819189.6 -819189.6
-819189.6
```

Best model:

Value of the maximum log-likelihood: -819189.6

step parameters:

	state 1	state 2	state 3	state 4
mean	2.680949	21.77835	382.0162	1945.086
sd	1.513453	19.21009	369.3465	1037.136

angle parameters:

	state 1	state 2	state 3	state 4
mean	3.1337626	-3.1320171	0.0389272	-0.03153871
concentration	0.3302889	0.3628805	0.1696356	1.55942826

5-state HMM:

Maximum log-likelihood values:

```
-817441.4 -817441.4 -817441.4 -817441.4 -817441.4 -817441.4 -817441.4
-817441.4 -817441.4 -817441.4 -817441.4 -817441.4 -817441.4 -817447.0
-817441.4
```

Best model:

Value of the maximum log-likelihood: -817441.4

step parameters:

	state 1	state 2	state 3	state 4	state 5
mean	2.524012	15.25322	220.6390	751.7033	2466.1359
sd	1.364508	12.17583	221.5885	484.0975	880.9561

angle parameters:

	state 1	state 2	state 3	state 4	state 5
mean	3.134879	-3.1259356	0.42186268	-0.008988095	-0.02347248
concentration	0.326152	0.3690583	0.01808188	0.531983987	1.80426823

Section S2. Choosing an appropriate number of states and model diagnostics

To check whether models with a different number of states change the model predictions, we fitted HMMs with three different sets of states (i.e., 3-state HMM, 4-state HMM, and 5-state HMM). We further checked whether the restrictions on the model transition probability matrix based on contemporaneous *in situ* observations showed more realistic decoded states predicted by the models. Thus, we first fitted an HMM without restrictions on the transition probability matrix for models with 3, 4, and 5 states. We second fitted an HMM with restrictions on the transition probability matrix for each state model. We then plotted a time series of step lengths colored by the decoded states from the models to check if both models provided similar decoded states (Figures S2 and S5). We observed that the decoded states were different from each other. We found that the 4-state model's decoded states were more reasonable than the 3- and 5-state models (Figures S2 and S5). For example, the constrained 4-state HMM correctly classified 33 of the 38 confirmed dead individuals, whereas the constrained 3 or 5-state HMM correctly classified fewer individuals than the 4-state HMM. Thus, we selected the 4-state model as our final model for the study.

We also plotted pseudo-residuals based on the estimated step lengths (top plots) and turning angles (bottom plots) from the 3-state HMM (Figure S4), the 4-state HMM (Figure S5) and 5-state HMM (Figure S6), and 4-state HHMM (Figure S7): (1). Time series plot of pseudo-residuals (left panel), (2). Q-Q plot of pseudo-residuals (middle panel), and (3) autocorrelation function (ACF) plot (right panel).

All models show similar trends from the time series plots, with residuals fluctuating randomly over time. We see some deviations in the lower tail of the step length distribution. The model predicts more very short step lengths than what is observed in the actual data. This may indicate the possibility of measurement error. Compared to the (H)HMMs, the 4-state constrained HMM appears to fit slightly better to the normal distribution, although some extreme values still deviate (Figure S5).

Section S3. 3-state HMMs

3.1. Unconstrained 3-state model

```
Value of the maximum log-likelihood: -837084.8

step parameters:
-----
              S              I              D
mean 1450.084 229.9058 15.61235
sd   1072.560 331.9978 20.55279

angle parameters:
-----
              S              I              D
mean      -0.04416724 2.98623791 -3.1334234
concentration 1.34392673 0.06907465 0.5466987

Regression coeffs for the transition probabilities:
-----
              1 -> 2          1 -> 3          2 -> 1          2 -> 3 3 -> 1
3 -> 2
(Intercept)      -0.9802697513 -6.6575337124 -2.4334206409 -1.5677767 -1e+06
-1e+06
shrub            -0.0152683936  0.0496176957 -0.0627233723 -0.4948396 -1e+06
-1e+06
time_since_release -0.0004278216  0.0008501638 -0.0003082179 -1.5771796 -1e+06
-1e+06

Transition probability matrix (based on mean covariate values):
-----
              S              I              D
S 0.75985182 0.2385520 1.596233e-03
I 0.05061246 0.9493875 1.278511e-113
D 0.00000000 0.0000000 1.000000e+00

Initial distribution:
-----
              S              I              D
3.864444e-11 1.000000e+00 1.188900e-16
```

3.2. Constrained 3-state model

```
Value of the maximum log-likelihood: -845574.2

step parameters:
-----
              S              I              D
mean  745.2912 294.7914  9.548312
```

```

sd    1087.3869 439.8388 11.125256

angle parameters:
-----
              S          I          D
mean          -0.01798164 0.711546828 2.913282
concentration  0.40192371 0.004641055 0.702596

Regression coeffs for the transition probabilities:
-----
              1 -> 2 1 -> 3 2 -> 1      2 -> 3 3 -> 1 3 -> 2
(Intercept)    -6.8968735682 -1e+06 -1e+06 -4.6815411 -1e+06 -1e+06
shrub           0.0533211177 -1e+06 -1e+06  0.2317948 -1e+06 -1e+06
time_since_release 0.0005727067 -1e+06 -1e+06 -9.4985281 -1e+06 -1e+06

Transition probability matrix (based on mean covariate values):
-----
              S          I D
S 0.9983791 0.001620881 0
I 0.0000000 1.000000000 0
D 0.0000000 0.000000000 1

Initial distribution:
-----
              S          I          D
6.761259e-01 3.238448e-01 2.925778e-05

```

3.3 Constrained 3-state model with known death (no convergence)

```

Value of the maximum log-likelihood: -1.797693e+308

step parameters:
-----
              S    I D
mean 1800 315 4
sd    1050 365 4

angle parameters:
-----
              S          I          D
mean          -0.03 0.07 3.141593
concentration  1.50 0.08 0.400000

Regression coeffs for the transition probabilities:
-----
              1 -> 2 1 -> 3 2 -> 1 2 -> 3 3 -> 1 3 -> 2
(Intercept)    -1.5 -1e+06 -1e+06    -1.5 -1e+06 -1e+06
shrub           0.0 -1e+06 -1e+06    0.0 -1e+06 -1e+06
time_since_release 0.0 -1e+06 -1e+06    0.0 -1e+06 -1e+06

```

Transition probability matrix (based on mean covariate values):

```
-----
          S          I          D
S 0.8175745 0.1824255 0.0000000
I 0.0000000 0.8175745 0.1824255
D 0.0000000 0.0000000 1.0000000
```

Initial distribution:

```
-----
          S          I          D
0.3333333 0.3333333 0.3333333
```

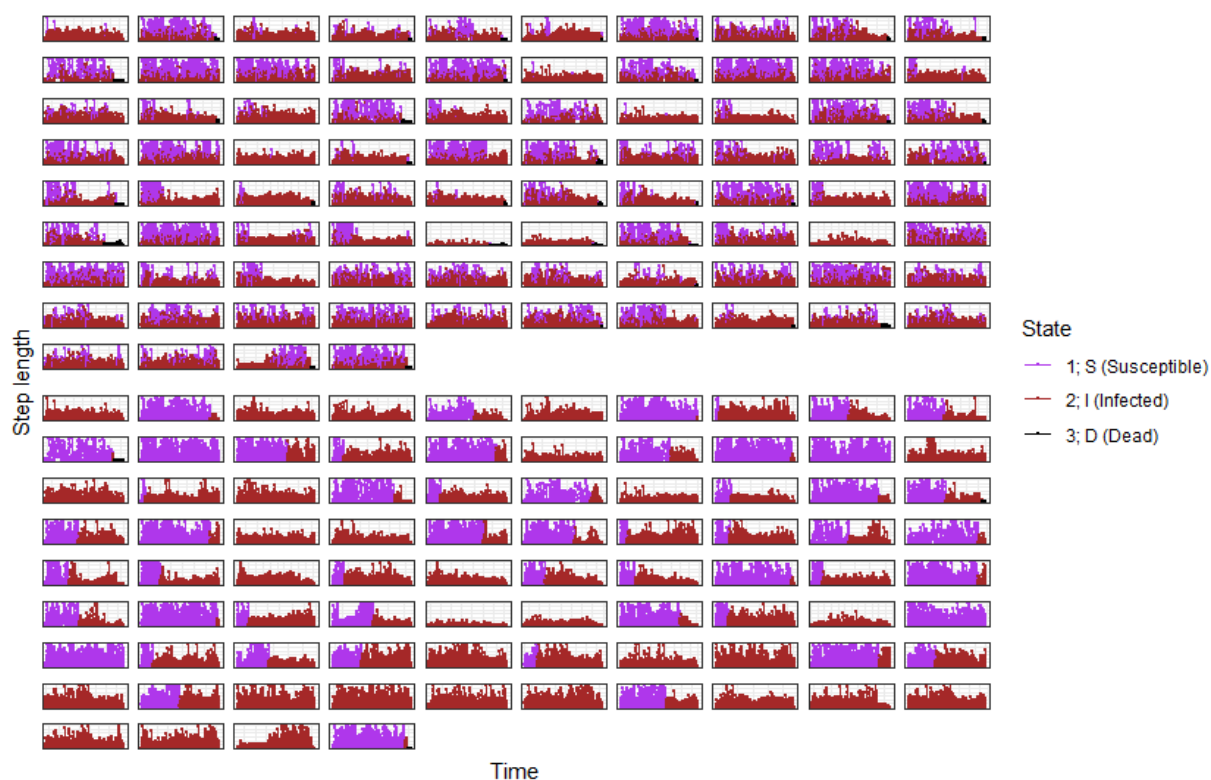


Figure S2. A time series of step lengths colored by the decoded states from the unconstrained 3-state HMM (a top panel) vs the constrained 3-state HMM (a bottom panel)

Section S4. 4-state (H)HMMs

4.1. Unconstrained 4-state model

```
Value of the maximum log-likelihood: -825583.4

step parameters:
-----
              SE          SR          I          D
mean 1802.752 317.9315 4.160769 8.466996
sd   1077.595 363.8234 3.086291 8.574073

angle parameters:
-----
              SE          SR          I          D
mean          -0.03640715 0.08437032 3.1408276 3.0658192
concentration  1.52331974 0.08402688 0.4032178 0.7689519

Regression coeffs for the transition probabilities:
-----
              1 -> 2      1 -> 3      1 -> 4      2 -> 1      2 -> 3
(Intercept)    -0.9506075888 -3.7405203961 -2.391813 -2.9138415733 -2.2063569020
shrub          -0.0264951454  0.0744572504 -4.895373 -0.0255400360  0.0248192845
time_since_release -0.0003633459 -0.0004985664 -6.008808 -0.0001662611 -0.0006136597
              2 -> 4      3 -> 1      3 -> 2      3 -> 4 4 -> 1 4 -> 2
(Intercept)    -1.774485 -2.7063824545 -0.030504019 -1.8038954963 -1e+06 -1e+06
shrub          -1.986169 -0.0733494976 -0.036589146 -0.6715370716 -1e+06 -1e+06
time_since_release -2.366991 -0.0009999114 -0.000153738  0.0005477064 -1e+06 -1e+06
              4 -> 3
(Intercept)    -1e+06
shrub          -1e+06
time_since_release -1e+06

Transition probability matrix (based on mean covariate values):
-----
              SE          SR          I          D
SE 0.74689275 0.2252484 0.02785886 0.000000e+00
SR 0.03784118 0.8597858 0.10237304 8.535314e-174
I  0.01907092 0.4130374 0.56705285 8.388240e-04
D  0.00000000 0.0000000 0.00000000 1.000000e+00

Initial distribution:
-----
              SE          SR          I          D
0.0114903337 0.0201674303 0.9673900647 0.0009521713
```

4.2. Constrained 4-state model

```
Value of the maximum log-likelihood: -835806.7
```

```
step parameters:
-----
              SE          SR          I          D
mean 1629.186 275.4228 194.0144 14.05722
sd   1094.530 391.6293 285.0358 18.09655
```

angle parameters:

	SE	SR	I	D
mean	-0.03771452	3.02589153	2.79016429	-3.1302463
concentration	1.55017641	0.03465011	0.04698539	0.5922272

Regression coeffs for the transition probabilities:

	1 -> 2	1 -> 3	1 -> 4	2 -> 1	2 -> 3	2 -> 4
(Intercept)	-0.9519603969	-1e+06	-1e+06	-2.7072246135	-6.762835566	-9.042585
shrub	-0.0122131328	-1e+06	-1e+06	-0.0214502286	-0.061725601	0.000000
time_since_release	-0.0002766136	-1e+06	-1e+06	0.0006709467	0.001667105	0.000000
	3 -> 1	3 -> 2	3 -> 4	4 -> 1	4 -> 2	4 -> 3
(Intercept)	-1e+06	-1e+06	-6.864947	-1e+06	-1e+06	-1e+06
shrub	-1e+06	-1e+06	0.000000	-1e+06	-1e+06	-1e+06
time_since_release	-1e+06	-1e+06	0.000000	-1e+06	-1e+06	-1e+06

Transition probability matrix (based on mean covariate values):

	SE	SR	I	D
SE	0.74724223	0.2527578	0.0000000000	0.0000000000
SR	0.05990208	0.9390728	0.0009140226	0.0001110592
I	0.00000000	0.0000000	0.9989573502	0.0010426498
D	0.00000000	0.0000000	0.0000000000	1.0000000000

Initial distribution:

	SE	SR	I	D
	2.558638e-06	9.472805e-01	5.271689e-02	4.163098e-09

Table S1. 38 confirmed dead individuals with their disease classifications and the constrained 4-state HMM prediction. The constrained 4-state HMM correctly predicted 33 dead individuals out of the 38 confirmed dead individuals.

ID	Disease classification	Model prediction
1	Dead	Infected
4	Dead	Dead
5	Dead	Dead
6	Dead	Dead
7	Dead	Dead
9	Dead	Dead
10	Dead	Dead
12	Dead	Dead
13	Dead	Dead
14	Dead	Dead
15	Dead	Dead
17	Dead	Dead
18	Dead	Dead
19	Dead	Dead
22	Dead	Dead
24	Dead	Dead
29	Dead	Dead
30	Dead	Dead
32	Dead	Susceptible
34	Dead	Infected
35	Dead	Dead
36	Dead	Dead
37	Dead	Infected
41	Dead	Dead
44	Dead	Infected

45	Dead	Dead
46	Dead	Dead
48	Dead	Dead
49	Dead	Infected
50	Dead	Dead
51	Dead	Dead
52	Dead	Dead
55	Dead	Dead
59	Dead	Dead
60	Dead	Dead
62	Dead	Dead
78	Dead	Dead
80	Dead	Dead

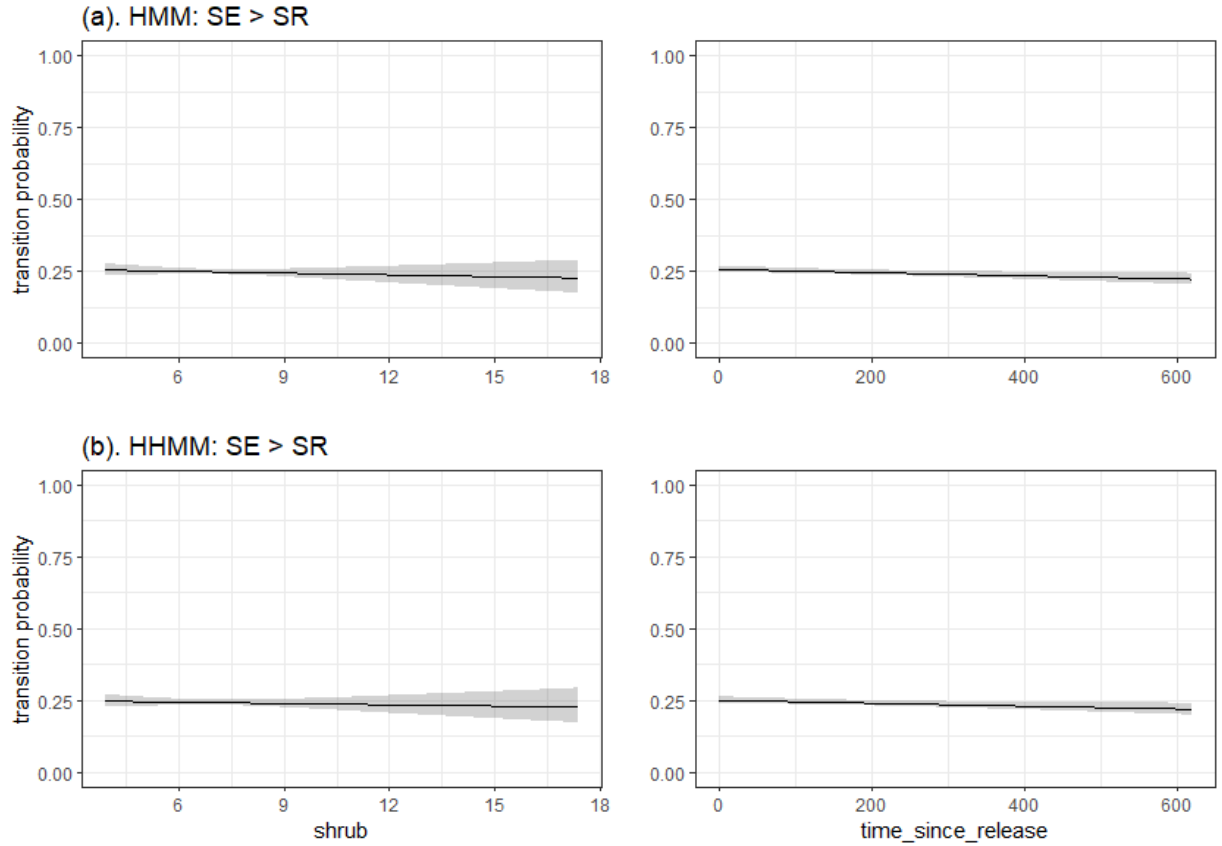


Figure S3. Transition probabilities from susceptible exploring (SE) to non-infected resting (SR) as a function of shrub cover and time since release with 95% confidence intervals from **(a)** the 4-state constrained HMM and **(b)** the 4-state constrained HHMM. For both models, there is no clear trend in the transition probabilities from exploring states to resting states as a function of shrub cover when oryx are susceptible (SE to SR). Similarly, there is no trend in the transition probabilities from exploring to resting states as a function of time since release when oryx are susceptible.

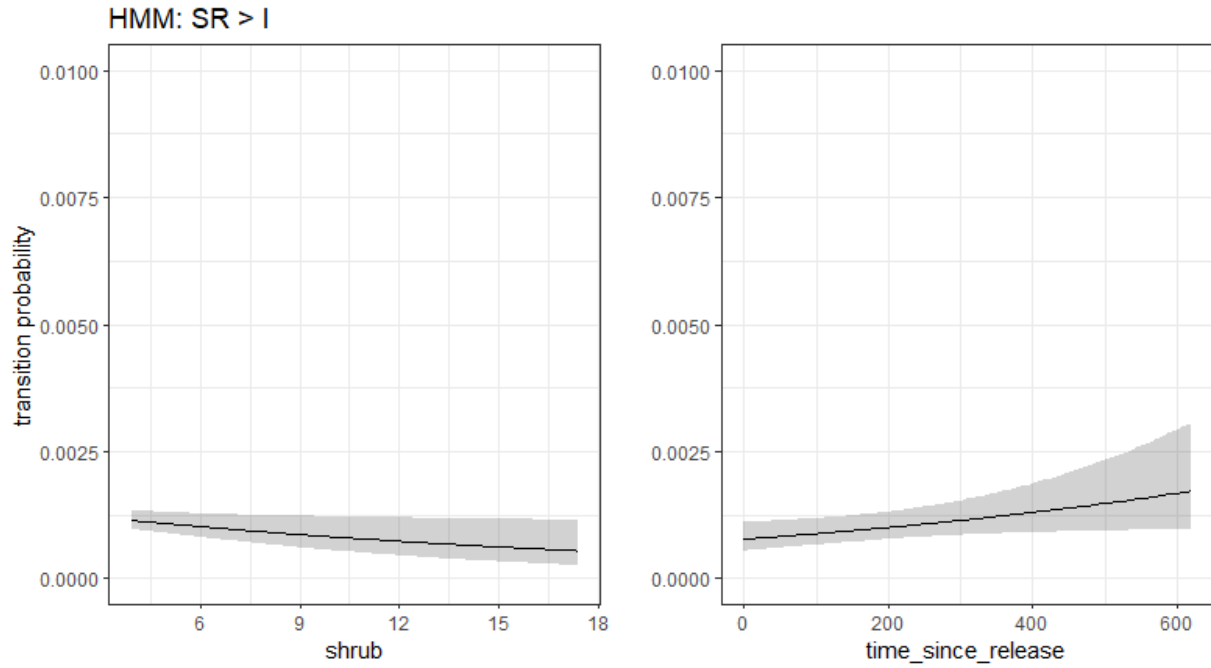


Figure S4. Transition probabilities from susceptible resting (SR) to infected (I) as a function of shrub cover and time since release with 95% confidence intervals from the 4-state constrained HMM. There is no trend in the transition probabilities from resting states to infected states as a function of shrub cover and time since release given the wide confidence bands. The y-axis is truncated to help with the visualization of small values.

Table S2. 38 confirmed dead individuals with their disease classifications and the constrained 4-state HHMM prediction. The constrained 4-state HHMM correctly predicted 27 dead individuals out of the 38 confirmed dead individuals

ID	Disease classification	Model prediction
1	Dead	Infected
4	Dead	Dead
5	Dead_Infected	Dead
6	Dead	Dead
7	Dead	Dead
9	Dead	Dead
10	Dead	Dead
12	Dead	Infected
13	Dead	Dead
14	Dead	Infected
15	Dead	Susceptible
17	Dead	Dead
18	Dead	Dead
19	Dead	Dead
22	Dead	Infected
24	Dead	Dead
29	Dead	Dead
30	Dead	Dead
32	Dead	Susceptible
34	Dead	Dead
35	Dead	Infected
36	Dead	Dead
37	Dead	Dead
41	Dead	Dead
44	Dead	Infected

45	Dead	Susceptible
46	Dead	Dead
48	Dead	Dead
49	Dead	Infected
50	Dead	Dead
51	Dead	Dead
52	Dead	Susceptible
55	Dead	Dead
59	Dead	Dead
60	Dead	Dead
62	Dead	Infected
78	Dead	Infected
80	Dead	Dead

4.3 Constrained 4-state model with known death (no convergence)

```
Value of the maximum log-likelihood: -1.797693e+308

step parameters:
-----
          SE  SR  I  D
mean 1800 380 25 4
sd   1050 370 20 4

angle parameters:
-----
          SE  SR  I  D
mean      -0.03 0.03 -3.00 3.141593
concentration 1.50 0.16 0.36 0.300000

Regression coeffs for the transition probabilities:
-----
          1 -> 2 1 -> 3 1 -> 4 2 -> 1 2 -> 3 2 -> 4 3 -> 1 3 -> 2 3 -> 4 4
-> 1 4 -> 2 4 -> 3
(Intercept)      -1.5 -1e+06 -1e+06      -1.5      -1.5      -1.5 -1e+06 -1e+06      -1.5
-1e+06 -1e+06 -1e+06
shrub              0.0 -1e+06 -1e+06      0.0      0.0      0.0 -1e+06 -1e+06      0.0
-1e+06 -1e+06 -1e+06
time_since_release 0.0 -1e+06 -1e+06      0.0      0.0      0.0 -1e+06 -1e+06      0.0
-1e+06 -1e+06 -1e+06

Transition probability matrix (based on mean covariate values):
-----
          SE  SR  I  D
SE 0.8175745 0.1824255 0.0000000 0.0000000
SR 0.1336597 0.5990210 0.1336597 0.1336597
I  0.0000000 0.0000000 0.8175745 0.1824255
D  0.0000000 0.0000000 0.0000000 1.0000000

Initial distribution:
-----
          SE  SR  I  D
0.25 0.25 0.25 0.25
```

4.4 Unconstrained 4-state hierarchical model

```
Value of the maximum log-likelihood: -815708.3

step parameters:
-----
          e  r  i  id  d  dd
mean 1691.204 297.8768 202.2163 202.2163 60.90892 60.90892
sd   1129.754 419.1183 293.2170 293.2170 97.55779 97.55779

angle parameters:
-----
          e  r  i  id  d  dd
mean      -0.03689851 3.135224 -3.13899047 -3.13899047 3.1367153 3.1367153
```

```
concentration
1.57803339 0.000000 0.06929608 0.06929608 0.2634562 0.2634562
```

```
-----
Regression coeffs for the transition probabilities:
-----
```

```
----- level1 -----
          1 -> 3    1 -> 5    3 -> 1    3 -> 5 5 -> 1 5 -> 3
I((level == "1") * 1) -2.175671 -4.901406 -1.956103 -2.862121 -1e+06 -1e+06
```

```
----- level2 -----
                                1 -> 2    2 -> 2
I((level == "2") * 1)          -0.9868733255 2.7683774306
I((level == "2") * shrub)      -0.0116017860 0.0025923825
I((level == "2") * time_since_release) -0.0003053025 -0.0004443198
```

```
                                3 -> 4 4 -> 4
I((level == "2") * 1)          -1e+06 -1e+06
I((level == "2") * shrub)      -1e+06 -1e+06
I((level == "2") * time_since_release) -1e+06 -1e+06
```

```
                                5 -> 6 6 -> 6
I((level == "2") * 1)          -1e+06 -1e+06
I((level == "2") * shrub)      -1e+06 -1e+06
I((level == "2") * time_since_release) -1e+06 -1e+06
```

```
-----
Transition probability matrix (based on mean covariate values):
-----
```

```
----- level1 -----
          nonInfected  Infected  Death
nonInfected 0.8920861 0.1012803 0.006633662
Infected    0.1179823 0.8343374 0.047680219
Death       0.0000000 0.0000000 1.000000000
```

```
----- level2 -----
          e          r
e 0.75382820 0.2461718
r 0.06208636 0.9379136
```

```
          i id
i 1 0
id 1 0
```

```
          d dd
d 1 0
dd 1 0
```

```
-----
Regression coeffs for the initial distribution:
-----
```



```

----- level1 -----
              state 3    state 5
(Intercept) -14453.31 -18613.09

----- level2 -----
              state 2
I((level == "2i") * 1) 1.630522

              state 4
I((level == "2i") * 1) 0

              state 6
I((level == "2i") * 1) 0
-----

```

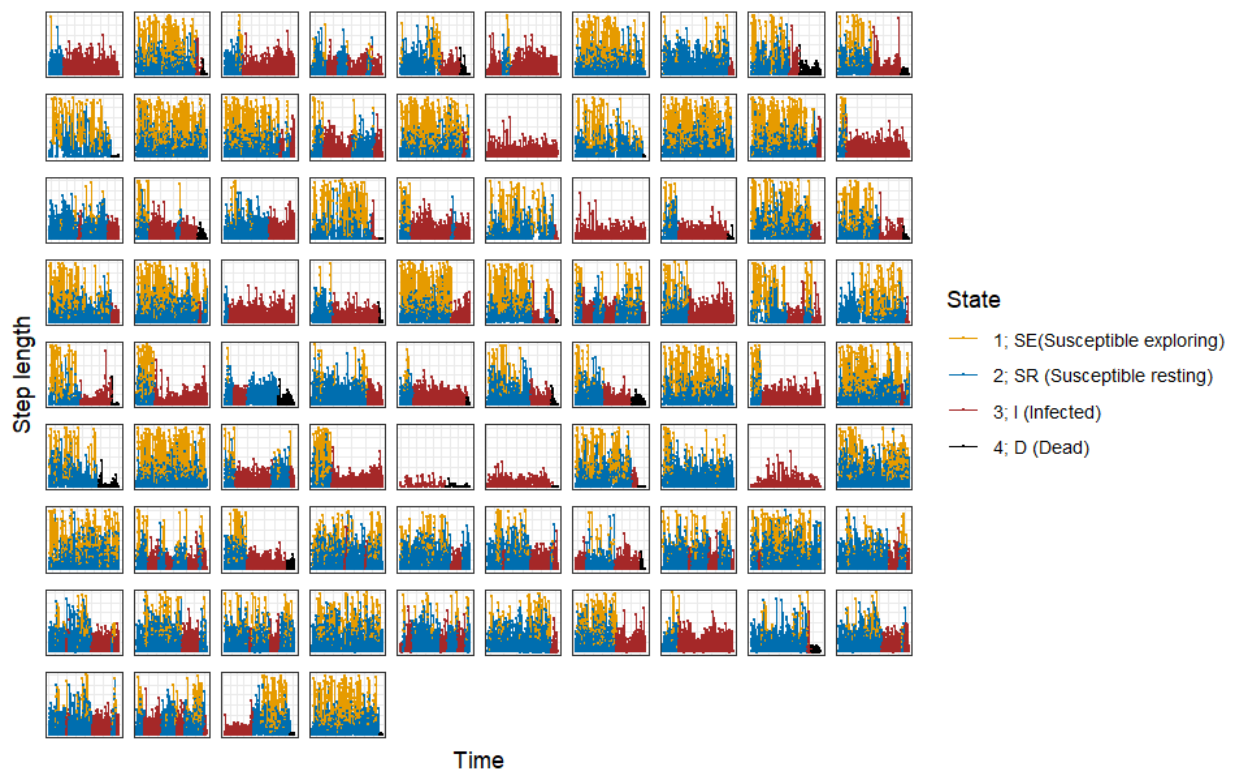


Figure S5. A time series of step lengths colored by the decoded states from the unconstrained 4-state HHMM

4.5 Constrained 4-state hierarchical model

Value of the maximum log-likelihood: -816070.8

step parameters:

```
-----
              e          r          i          id          d          dd
mean 1675.549 283.8440 217.3480 217.3480 64.90456 64.90456
sd   1128.406 402.6501 313.2039 313.2039 104.97910 104.97910
```

angle parameters:

```
-----
              e          r          i          id          d
dd
mean          -0.03434989 2.381206e+00 -3.00058057 -3.00058057 3.0170807
3.0170807
concentration 1.54178840 9.608558e-10 0.04483596 0.04483596 0.2640674
0.2640674
```

Regression coeffs for the transition probabilities:

```
----- level1 -----
              1 -> 3      1 -> 5 3 -> 1      3 -> 5 5 -> 1 5 -> 3
I((level == "1") * 1) -2.899543 -4.812639 -1e+06 -2.917137 -1e+06 -1e+06
```

```
----- level2 -----
              1 -> 2      2 -> 2
I((level == "2") * 1)          -1.0317888207 2.7318502453
I((level == "2") * shrub)      -0.0082644265 0.0221857338
I((level == "2") * time_since_release) -0.0002807144 -0.0009640931
```

```
              3 -> 4 4 -> 4
I((level == "2") * 1)          -1e+06 -1e+06
I((level == "2") * shrub)      -1e+06 -1e+06
I((level == "2") * time_since_release) -1e+06 -1e+06
```

```
              5 -> 6 6 -> 6
I((level == "2") * 1)          -1e+06 -1e+06
I((level == "2") * shrub)      -1e+06 -1e+06
I((level == "2") * time_since_release) -1e+06 -1e+06
```

Transition probability matrix (based on mean covariate values):

```
----- level1 -----
nonInfected  Infected  Death
nonInfected 0.9405791 0.05177737 0.007643511
Infected    0.0000000 0.94868713 0.051312872
Death       0.0000000 0.00000000 1.000000000
```

```

----- level2 -----
              e          r
e 0.75698486 0.2430151
r 0.06095609 0.9390439

      i id
i 1 0
id 1 0

      d dd
d 1 0
dd 1 0

-----

-----
Regression coeffs for the initial distribution:
-----
----- level1 -----
              state 3    state 5
(Intercept) -52.42732 -71.61341

----- level2 -----
              state 2
I((level == "2i") * 1) 1.669756

              state 4
I((level == "2i") * 1) 2.959402e-09

              state 6
I((level == "2i") * 1) 2.959383e-09
-----

```

Section S5. 5-state HMMs

5.1. Unconstrained 5-state model

```
Value of the maximum log-likelihood: -822342.8

step parameters:
-----
              SE      SR      IE      IR      D
mean 1959.649 383.6203 21.26635 2.658252 2.1129006
sd   1052.144 372.7165 18.93075 1.490357 0.5262408

angle parameters:
-----
              SE      SR      IE      IR      D
mean          -0.03113489 0.04102629 -3.1293622 3.1312131 2.631227
concentration  1.56198632 0.16905466  0.3584423 0.3339477 6.648088

Regression coeffs for the transition probabilities:
-----
              1 -> 2      1 -> 3      1 -> 4      1 -> 5      2 ->
1      2 -> 3      2 -> 4      2 -> 5      3 -> 1
(Intercept)          -1.0162908447 -2.685773e+00 -2.990627 -1.834031
-2.878513e+00 -2.187746441 -2.3379370164 -24.96851 -3.226942e+00
shrub              -0.0219192633  1.976282e-03 -6.022947 -2.349271
-2.943447e-02 -0.006404594  0.0247462881  0.00000 -2.812501e-02
time_since_release -0.0004811317 -1.513205e-05 -1.164282 -1.349730
-3.232238e-05  0.000378894 -0.0008349726  0.00000  8.185061e-05
              3 -> 2      3 -> 4      3 -> 5      4 -> 1      4
-> 2      4 -> 3      4 -> 5 5 -> 1 5 -> 2 5 -> 3 5 -> 4
(Intercept)          -0.1205061713 -1.357008588 -1.2829509 -3.010103382
-0.0771407536 -1.5728257701 -1.786558 -1e+06 -1e+06 -1e+06 -1e+06
shrub              -0.0307891633  0.007738664 -0.5289330 -0.039744950
-0.0278891124 -0.0249301076 -2.123231 -1e+06 -1e+06 -1e+06 -1e+06
time_since_release -0.0003757599 -0.001686547 -0.6775713 -0.001417873
-0.0002530488  0.0005630142 -1.397285 -1e+06 -1e+06 -1e+06 -1e+06

Transition probability matrix (based on mean covariate values):
-----
              SE      SR      IE      IR      D
SE 0.73786075 0.2112493 0.05088998 1.529051e-102 1.206953e-103
SR 0.03596572 0.7937169 0.09041440 7.990293e-02 1.137577e-11
IE 0.01722097 0.3505810 0.52369601 1.085021e-01 9.591827e-51
IR 0.01514746 0.3738021 0.09761470 5.134357e-01 2.042947e-106
D 0.00000000 0.0000000 0.00000000 0.000000e+00 1.000000e+00

Initial distribution:
-----
              SE      SR      IE      IR      D
6.528100e-04 1.267492e-04 6.851729e-01 3.140378e-01 9.794411e-06
```

5.2. Constrained 5-state model

Value of the maximum log-likelihood: -829127.5

step parameters:

```
-----
              SE          SR          IE          IR          D
mean 1891.260 296.2557 379.8396 4.315509 5.820463
sd   1185.073 414.9153 454.7442 3.261060 5.241028
```

angle parameters:

```
-----
              SE          SR          IE          IR          D
mean          -0.0334418 1.823797672 0.02155107 3.1385940 3.1092579
concentration  1.8121634 0.008826148 0.13890107 0.4392679 0.8135199
```

Regression coeffs for the transition probabilities:

```
-----
              1 -> 2 1 -> 3 1 -> 4 1 -> 5          2 -> 1 2 -> 3          2
-> 4 2 -> 5 3 -> 1 3 -> 2          3 -> 4 3 -> 5 4 -> 1
(Intercept)    -9.658813e-01 -1e+06 -1e+06 -1e+06 -2.3667863026 -1e+06
-6.6977437749 -14.263 -1e+06 -1e+06 -2.3143799222 -1e+06 -1e+06
shrub          -2.603151e-02 -1e+06 -1e+06 -1e+06  0.0024873074 -1e+06
0.1120207665   0.000 -1e+06 -1e+06  0.0294247581 -1e+06 -1e+06
time_since_release -6.494087e-05 -1e+06 -1e+06 -1e+06  0.0005266694 -1e+06
0.0002775426   0.000 -1e+06 -1e+06 -0.0003619823 -1e+06 -1e+06
              4 -> 2          4 -> 3          4 -> 5 5 -> 1 5 -> 2 5 -> 3 5 -> 4
(Intercept)    -1e+06 -0.1603475754 -6.231782 -1e+06 -1e+06 -1e+06 -1e+06
shrub          -1e+06 -0.0264425082  0.000000 -1e+06 -1e+06 -1e+06 -1e+06
time_since_release -1e+06 -0.0002612298  0.000000 -1e+06 -1e+06 -1e+06 -1e+06
```

Transition probability matrix (based on mean covariate values):

```
-----
              SE          SR          IE          IR          D
SE 0.76178139 0.2382186 0.00000000 0.000000000 0.0000000e+00
SR 0.09391075 0.9034921 0.00000000 0.002596575 5.775399e-07
IE 0.00000000 0.00000000 0.8968311 0.103168870 0.0000000e+00
IR 0.00000000 0.00000000 0.4028632 0.595965214 1.171635e-03
D  0.00000000 0.00000000 0.00000000 0.000000000 1.0000000e+00
```

Initial distribution:

```
-----
              SE          SR          IE          IR          D
0.0092222272 0.7275775009 0.0263061181 0.2367375727 0.0001565811
```

5.3 Constrained 5-state model with known death (no convergence)

Value of the maximum log-likelihood: -1.797693e+308

step parameters:

```
-----
              SE  SR  IE  IR  D
mean 1800 750 220 15 2.5
sd   1050 484 220 12 1.0
```

angle parameters:

```
-----
```

	SE	SR	IE	IR	D
mean	-0.03	-0.009	0.40	-3.1	3.141593
concentration	1.80	0.530	0.01	0.3	0.300000

Regression coeffs for the transition probabilities:

```

-----
1 -> 2 1 -> 3 1 -> 4 1 -> 5 2 -> 1 2 -> 3 2 -> 4 2 -> 5 3 -> 1 3
-> 2 3 -> 4 3 -> 5 4 -> 1 4 -> 2 4 -> 3 4 -> 5 5 -> 1
(Intercept)      -1.5 -1e+06 -1e+06 -1e+06      -1.5 -1e+06      -1.5      -1.5 -1e+06
-1e+06      -1.5 -1e+06 -1e+06 -1e+06      -1.5      -1.5 -1e+06
shrub          0.0 -1e+06 -1e+06 -1e+06          0.0 -1e+06          0.0          0.0 -1e+06
-1e+06          0.0 -1e+06 -1e+06 -1e+06          0.0          0.0 -1e+06
time_since_release 0.0 -1e+06 -1e+06 -1e+06          0.0 -1e+06          0.0          0.0 -1e+06
-1e+06          0.0 -1e+06 -1e+06 -1e+06          0.0          0.0 -1e+06
5 -> 2 5 -> 3 5 -> 4
(Intercept)      -1e+06 -1e+06 -1e+06
shrub            -1e+06 -1e+06 -1e+06
time_since_release -1e+06 -1e+06 -1e+06

```

Transition probability matrix (based on mean covariate values):

```

-----
SE      SR      IE      IR      D
SE 0.8175745 0.1824255 0.0000000 0.0000000 0.0000000
SR 0.1336597 0.5990210 0.0000000 0.1336597 0.1336597
IE 0.0000000 0.0000000 0.8175745 0.1824255 0.0000000
IR 0.0000000 0.0000000 0.1542808 0.6914385 0.1542808
D 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000

```

Initial distribution:

```

-----
SE SR IE IR D
0.2 0.2 0.2 0.2 0.2

```



Figure S6. A time series of step lengths colored by the decoded states from the unconstrained 5-state HMM (a top panel) vs the constrained 5-state HMM (a bottom panel)

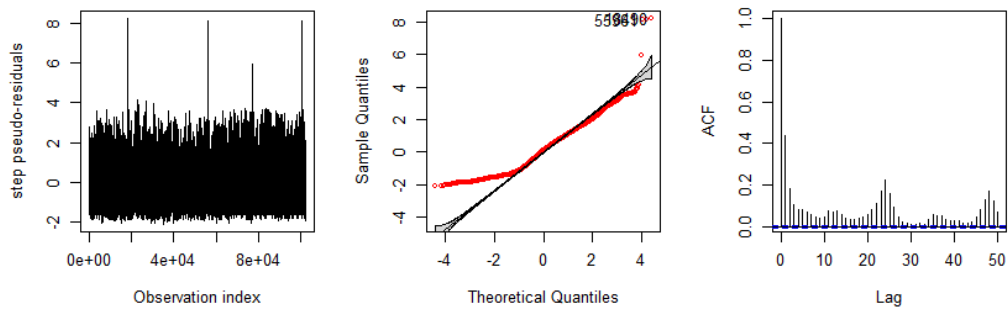


Figure S7. Pseudo-residual plot for the constrained 3-state HMM.

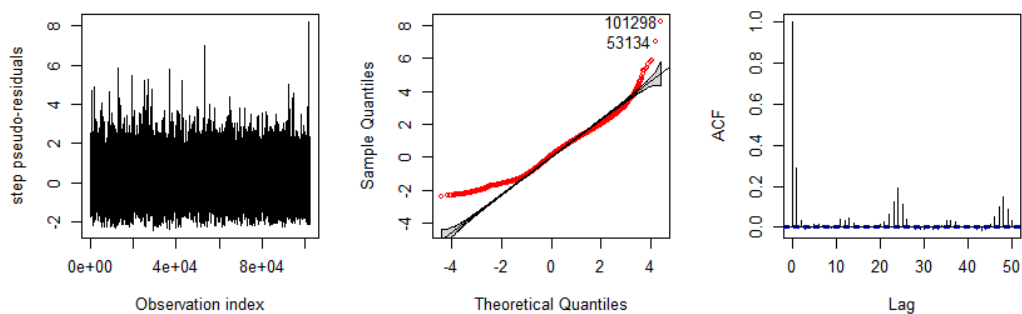


Figure S8. Pseudo-residual plot for the constrained 4-state HMM.

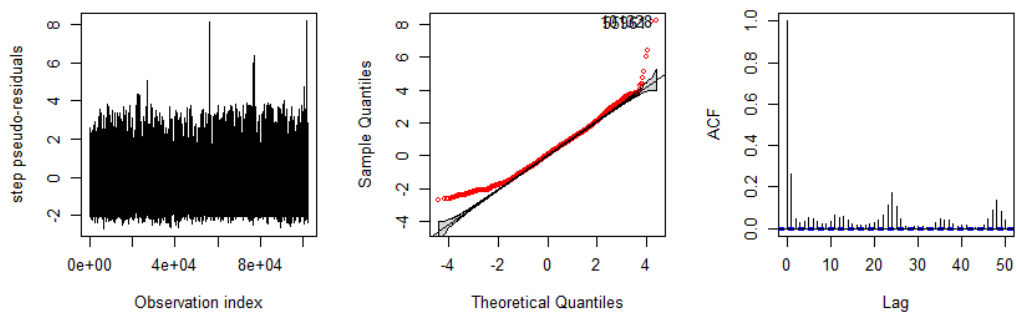


Figure S9. Pseudo-residual plot for the constrained 5-state HMM.

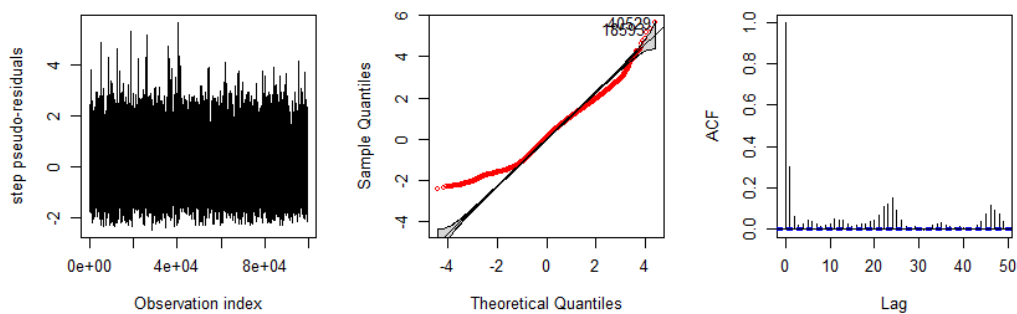


Figure S10. Pseudo-residual plot for the constrained 4-state HHMM.